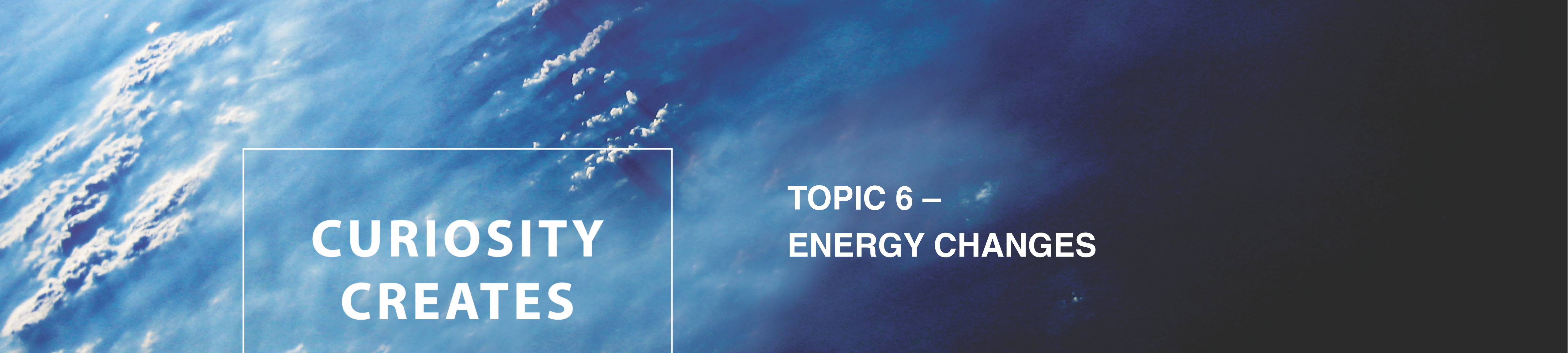




**CURIOSITY
CREATES**

**A BETTER
WORLD**

**TOPIC 6 –
ENERGY CHANGES**

QEng Prep - Chemistry



FACULTY OF
ENGINEERING AND
APPLIED SCIENCE

KEY CONCEPTS

The key concepts covered in this lecture include:

- Solving problems involving analysis of heat transfer in a chemical reaction, using the equation $Q = mc\Delta T$;
- Solving problems related to energy changes in a chemical reaction, using Hess's law; and
- Calculating the heat of reaction for a formation reaction, using a table of standard enthalpies of formation and applying Hess's law.

HEAT TRANSFER



INTRODUCTION

First Law of Thermodynamics

- Energy can be neither created nor destroyed but it can be converted from one form to another.

Energy Changes

- All chemical reactions are accompanied by some form of energy change; changes can be very obvious (e.g. coal burning) or can go unnoticed.
- There are two types of energy changes:
 - **Exothermic:** energy (in the form of heat) is released
 - **Endothermic:** energy (in the form of heat) is absorbed

CALCULATING HEAT TRANSFER

Heat

- Heat, q , is thermal energy transferred from a hotter system to a cooler system that are in contact with each other.
- Temperature is a measure of the average kinetic energy of the atoms or molecules in the system.
- We can calculate the heat released or absorbed by using the specific heat capacity, C , the mass of the substance, m , and the change in temperature, ΔT , in the following equation:

$$q = m \times C \times \Delta T$$

Where:

$$\Delta T = T_{\text{final}} - T_{\text{initial}}$$

- Recall also that the heat lost by one entity is absorbed by another entity i.e. $q_A = -q_B$
-

CALCULATING HEAT TRANSFER

Examples:

1. How much energy is required to heat 50 grams of water from 20°C to 85°C? The specific heat capacity of water is 4.184 J/g°C.
2. 4000J of heat energy is applied to a 70g sample of water initially measured to be at 35°C. What is the final temperature of the water?
3. 150 grams of an unknown metal at 150°C was dropped into a bucket containing 50 g of water initially at 21°C. The final temperature of the mixture was 35°C. What is the specific heat capacity of the unknown metal?

CALCULATING HEAT TRANSFER

Examples:

1. How much energy is required to heat 50 grams of water from 20°C to 85°C? The specific heat capacity of water is 4.184 J/g°C.

CALCULATING HEAT TRANSFER

Examples (continued):

2. 4000J of heat energy is applied to a 70g sample of water initially measured to be at 35°C. What is the final temperature of the water?

CALCULATING HEAT TRANSFER

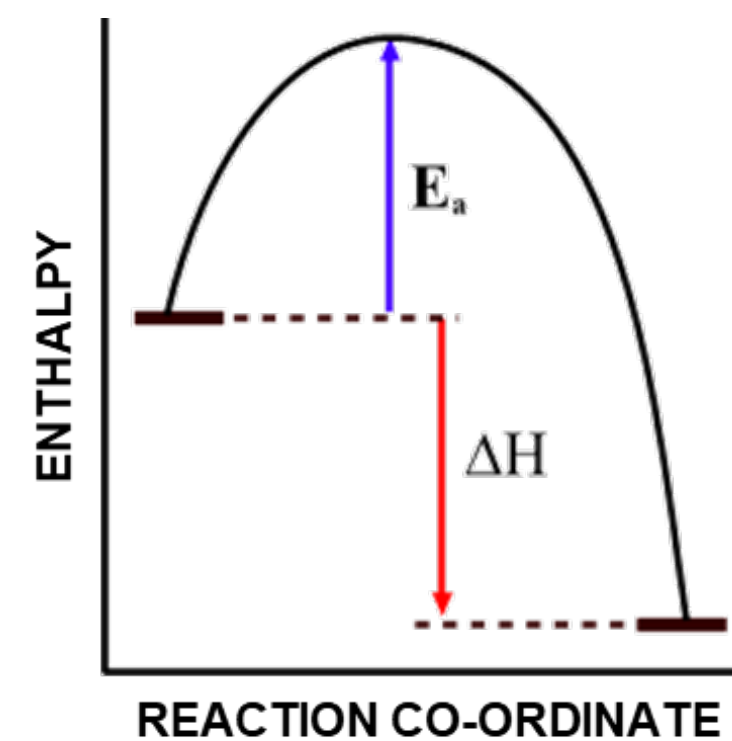
Examples (continued):

3. 150 grams of an unknown metal at 150°C was dropped into a bucket containing 50 g of water initially at 21°C . The final temperature of the mixture was 35°C . What is the specific heat capacity of the unknown metal?

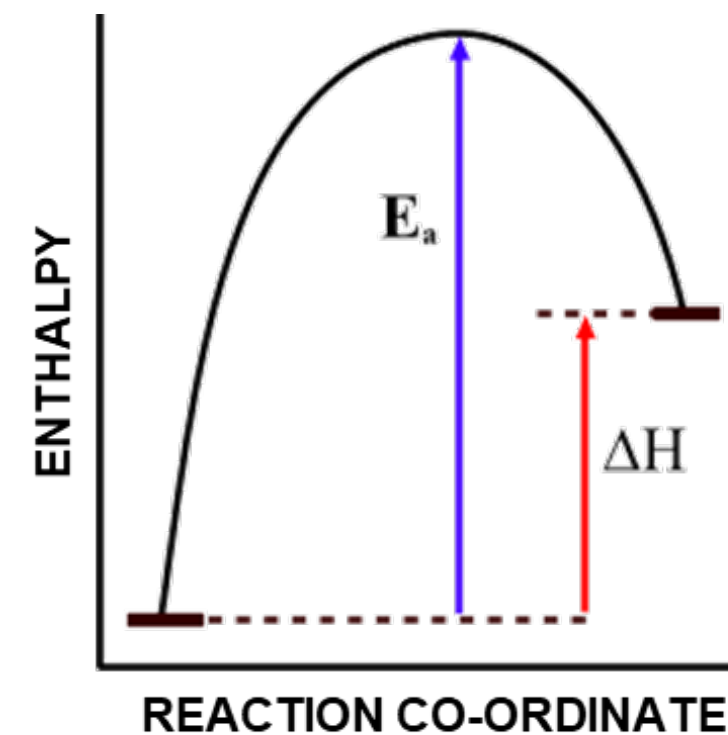
ENTHALPY

Definition

- Enthalpy, ΔH , is a measure of the heat content of a substance at constant pressure. You cannot measure the actual enthalpy of a substance, you can only measure an enthalpy change. By definition: $\Delta H = \Delta H_{\text{products}} - \Delta H_{\text{reactants}}$
- If $H_{\text{reactants}} > H_{\text{products}}$, then ΔH is negative and the reaction is exothermic (heat is released):



- If $H_{\text{reactants}} < H_{\text{products}}$, then ΔH is positive and the reaction is endothermic (heat is absorbed):



ENTHALPY

Standard Enthalpy Changes

- Enthalpy values vary according to the conditions
- A substance under the following conditions is said to be in its standard state:
 - Pressure: 100 kPa (1 atm)
 - Temperature: usually 298K (25°C)
- Remember to assign the correct subscript, (g), (l) or (s) to indicate which state a substance is in
- To tell if standard conditions are used, we modify the symbol for ΔH to ΔH°_{298}

ENTHALPY

Standard Enthalpy of Formation (ΔH°_f)

- Defined as the enthalpy change when one mole of a compound is formed in its standard state from its elements in their standard states. For example, the standard enthalpy of formation of $\text{CO}_2(\text{g})$ is -393.5 kJ/mol .
- Starting with the reactants at a pressure of 1 atm and 25°C (with the carbon present as graphite, the most stable form of carbon under these conditions) and ending with one mole of CO_2 , also at 1 atm and 25°C , this is the enthalpy change for the exothermic reaction:



- For nitrogen dioxide, $\text{NO}_2(\text{g})$, ΔH°_f is 33.2 kJ/mol . This is the enthalpy change for the reaction:



- Notice that there is only one mole of product on the right-hand side of each equation i.e., a reaction equation with $1/2$ mole of N_2 and 1 mole of O_2 is correct in this case because the standard enthalpy of formation always refers to 1 mole of product, $\text{NO}_2(\text{g})$.

STANDARD ENTHALPY OF FORMATION

- A excerpt of the table of standard enthalpies of formation of many common substances in shown below. By definition, the standard enthalpy of formation of an element in its most stable form is equal to zero under standard conditions, which is 1 atm for gases and 1 M for solutions.

Standard Thermodynamic Properties for Selected Substances	
Substance	ΔH_f° (kJ mol ⁻¹)
aluminum	
Al(s)	0
Al(g)	324.4
Al ³⁺ (aq)	-531
Al ₂ O ₃ (s)	-1676
AlF ₃ (s)	-1510.4
AlCl ₃ (s)	-704.2
AlCl ₃ ·6H ₂ O(s)	-2691.57
Al ₂ S ₃ (s)	-724.0
Al ₂ (SO ₄) ₃ (s)	-3445.06

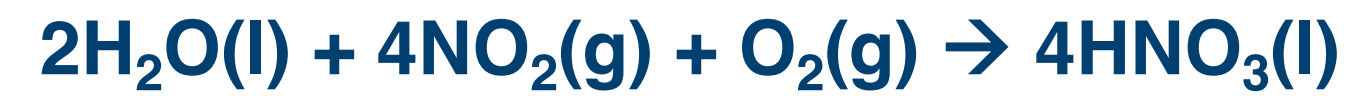
STANDARD ENTHALPY OF FORMATION

- By definition, the standard enthalpy of formation for a substance can be calculated using the following equation:

$$\Delta H_{\text{rxn}} = \sum \Delta H_f^\circ(\text{products}) - \sum \Delta H_f^\circ(\text{reactants})$$

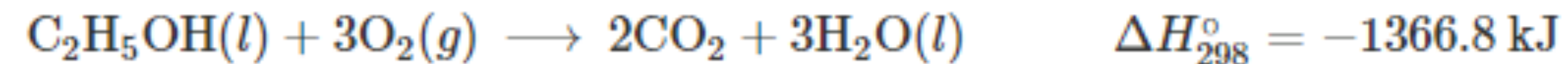
Example:

Calculate the standard enthalpy change for the following reaction, given that the standard enthalpies of formation of water, nitrogen dioxide and nitric acid are -286, +33 and -173 kJ/mol respectively:



ENTHALPY OF COMBUSTION

- Standard enthalpy of combustion (ΔH°_c) is the enthalpy change when 1 mole of a substance burns (i.e., combines vigorously with oxygen) under standard state conditions; it is sometimes called “heat of combustion.”
- For example, the enthalpy of combustion of ethanol, -1366.8 kJ/mol, is the amount of heat produced when one mole of ethanol undergoes complete combustion at 25°C and 1 atmosphere pressure, yielding products also at 25°C and 1 atm:



- Similarly to the enthalpy of formation, the enthalpy of combustion can be calculated if we have information regarding the reactants and products of a combustion reaction:

$$\Delta H = \sum \Delta H^\circ_c(\text{products}) - \sum \Delta H^\circ_c(\text{reactants})$$

ENTHALPY OF COMBUSTION

Example:

Calculate the standard enthalpy of formation of methane; the standard enthalpies of combustion of carbon, hydrogen and methane are -394, -286 and -890 kJ/mol.



HESS'S LAW

- There are two ways to determine the amount of heat involved in a chemical change:
 - Measure it experimentally; or
 - Calculate it from other experimentally determined enthalpy changes.
- This type of calculation usually involves the use of Hess's law, which states:

“If a process can be written as the sum of several stepwise processes, the enthalpy change of the total process equals the sum of the enthalpy changes of the various steps.”

- Hess's law is valid because enthalpy changes depend only on where a chemical process starts and ends, but not on the path it takes from start to finish.
-

HESS'S LAW

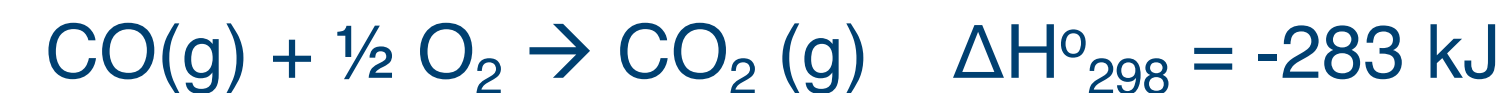
- For example, we can think of the reaction of carbon with oxygen to form carbon dioxide as occurring either directly or by a two-step process. The direct process is written:



In the two-step process, first carbon monoxide is formed:



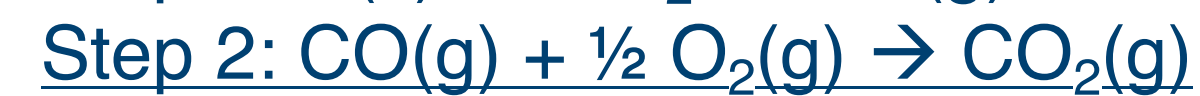
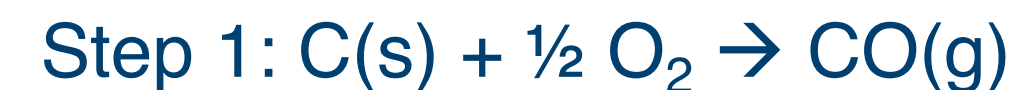
Then, carbon monoxide reacts further to form carbon dioxide:



By inspection, we see that ΔH of the overall reaction is the same whether it occurs in one step or two. This finding (overall ΔH for the reaction = sum of ΔH values for reaction “steps” in the overall reaction) is true in general for chemical and physical processes.

HESS'S LAW

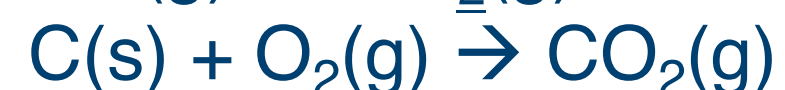
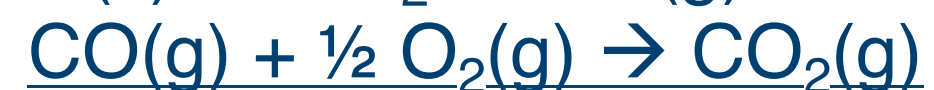
If we continue with our example of the reaction of carbon with oxygen to form carbon dioxide, the equation describing the **overall reaction is the sum** of these two chemical changes:



Because the CO produced in Step 1 is consumed in Step 2, the net change is:



According to Hess's law, the **enthalpy change of the reaction will equal the sum of the enthalpy changes of the steps**. We can apply the data from the experimental enthalpies of combustion to find the enthalpy change of the entire reaction from its two steps:

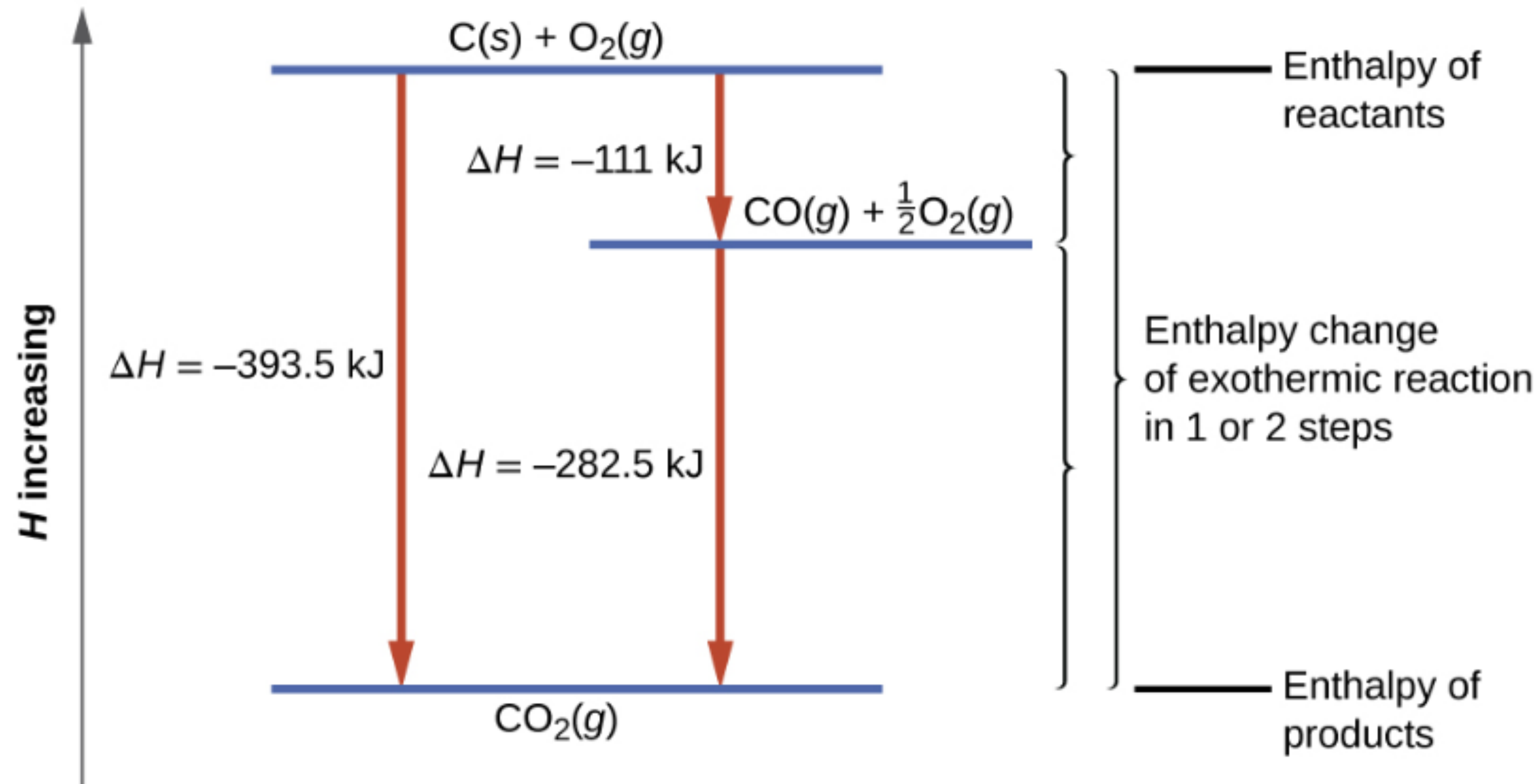


$$\Delta H^\circ_{298} = -111 \text{ kJ}$$

$$\Delta H^\circ_{298} = -283 \text{ kJ}$$

$$\Delta H^\circ_{298} = -394 \text{ kJ}$$

HESS'S LAW

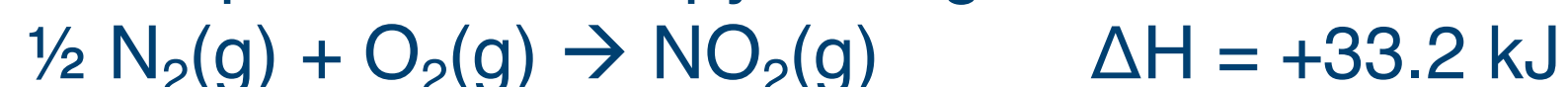


HESS'S LAW

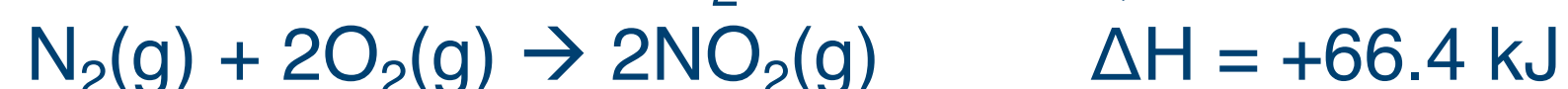
- Before we further practice using Hess's law, let us recall two important features of ΔH :

1. ΔH is directly proportional to the quantities of reactants or products. In general, if we multiply or divide an equation by a number, then the enthalpy change should also be multiplied or divided by the same number.

- For example, the enthalpy change for the reaction forming **1** mole of $\text{NO}_2(\text{g})$ is +33.2 kJ:



When **2** moles of NO_2 are formed, the ΔH will be twice as large:



2. ΔH for a reaction in one direction is equal in magnitude and opposite in sign to ΔH for the reaction in the reverse direction.

- For example, given that:



Then, for the “reverse” reaction, the enthalpy change is also “reversed”:



HESS'S LAW

Examples:

1. Determine the enthalpy of formation, ΔH°_f , of $\text{FeCl}_3(\text{s})$ from the enthalpy changes of the following two-step process that occurs under standard state conditions:



2. At 25°C, the following heats of reaction are known:



At the same temperature, use Hess's law to calculate $\Delta H^\circ_{\text{rxn}}$ for the reaction:



HESS'S LAW

Example:

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Examples:

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REFERENCES

1. Chemistry: Atoms First, an OpenStax resource (<https://cnx.org/contents/RTmulxzM@7.152:9H-cauk3@13/Preface>)